

Simulating projective Augmented Reality Visualizations in Virtual Reality: Is VR a feasible Environment for medical AR Evaluations?

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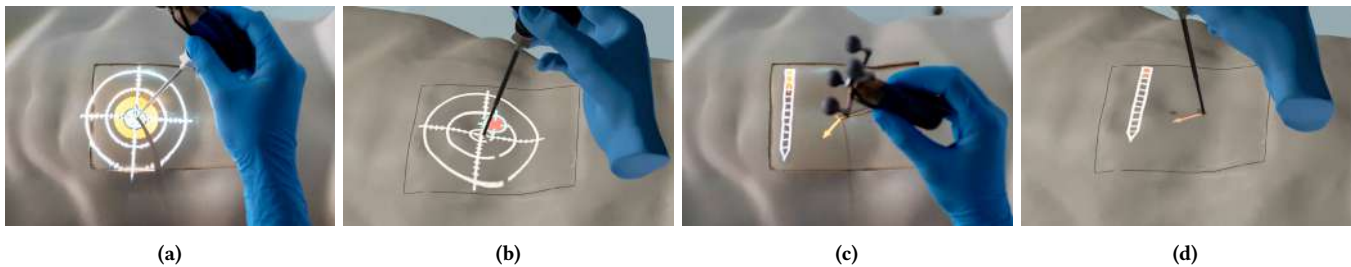


Figure 1: Compared needle navigation visualizations using projective AR and the respective simulation in VR. Left two images show the Cross concept while the right two figures represent the Arrow visualization. The respective left image shows photographed projections and the right ones are screenshots from the VR environment.

ABSTRACT

Augmented Reality (AR) has demonstrated potential in medical applications, such as enhancing surgical navigation. However, evaluating medical AR visualizations entails high costs and effort to provide suitable hardware solutions. This is particularly crucial in projective AR, as these systems require several error-prone calibration and registration steps. This work investigates the suitability of Virtual Reality (VR) as a cost-effective and controlled study environment for evaluating projective AR visualizations. A virtual twin of a real laboratory environment was created, and a user study comparing two needle navigation visualizations was conducted. The study simulated identical experiments in both AR and VR to assess if similar results would emerge. Our findings indicate that both AR and VR experiments exhibited comparable effects in terms of performance and workload of both needle insertion visualizations. This study serves as a preliminary step in demonstrating the feasibility of using VR as an evaluation environment for projective AR visualizations.

CCS CONCEPTS

• **Computing methodologies** → **Mixed / augmented reality**;
• **Applied computing** → **Health informatics**; • **Hardware** → **Displays and imagers**; • **Human-centered computing** → **Empirical studies in visualization**.

KEYWORDS

Medical Augmented Reality, Virtual Reality, Surgical Navigation

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1 INTRODUCTION

The potential of Augmented Reality (AR) has been shown in numerous medical applications, providing innovative solutions for medical training, visualization, and patient care [7, 10]. Especially during minimally invasive interventions, the direct visualization of virtual navigation information onto the patient's anatomy has the potential to improve the precision and efficiency [25]. Although several AR visualizations to support the physician have been developed and tested by researchers [3, 6, 13], the practical implementation and

evaluation of medical AR visualizations remain challenging and limited in clinical settings due to the dynamic and context-dependent nature of medical procedures. This is especially true for projective AR, also referred to as spatial AR in the literature. This modality enhances physical objects using video projectors. In medical settings, those projectors can be spatially separated from the physician, thus causing less sterility issues and discomfort compared to hand-held or head-worn devices. Other advantages compared to different AR modalities, e.g., optical see-through headsets, include reduced cognitive costs and the potential of increased performance [2, 14]. However, the setup, calibration and spatial registration of such systems are comparably expensive, time-consuming and error-prone, which, despite the shown benefits, has contributed to the limited use of projective AR in the medical domain. Furthermore, uncertainties caused by tracking and registration errors negatively affect usability evaluations of AR contents because performance measures may be biased by those inaccuracies.

In this context, virtual reality (VR) enables the rapid creation of immersive environments that can simulate real-world scenarios with high fidelity [18, 20]. The application of VR in different research areas has demonstrated its potential to provide controlled and standardized evaluation environments [17, 23, 30]. In the medical domain, VR was used to negate effects of registration errors for the evaluation of laparoscopic AR visualizations [15], and as a realistic and immersive evaluation environment for the assessment of interaction techniques [31]. It was also used for the evaluation of AR visualizations in logistics [32] and airborne surveillance [1] to mitigate high logistical efforts of conventional testing, and in industrial maintenance to overcome limitations due to safety at deployment [4]. In contrast to conventional real-world scenarios, conducting empirical research in simulated VR environments has different advantages. Martin-Gomez et al. [26, 27] highlighted benefits of using VR as a controlled evaluation environment for optical see-through AR visualizations in terms of consistent geometrical properties of objects and diminished effects of external factors such as imprecise eye calibration, reduced field of view, or inaccurate tracking. Further identified advantages include saving of time and costs, as well as the potential of earlier and more efficient evaluations [28]. Furthermore, by leveraging these capabilities of VR, virtual environments (VEs) can be created to simulate different AR devices and to evaluate AR visualizations in a more robust and reproducible manner [22]. As such, VR was used to simulate AR displays for vehicle navigation [19], or hand-held video see-through AR [5, 35]. With respect to optical see-through AR head-mounted displays (HMDs), Lacoche et al. [22] could show that there were very few differences between an actual AR application and its simulation in VR. Moreover, simulating AR in VR has the potential of leveraging prototyping. This method was used for fast development and testing to decide whether further development and implementation of the new AR application are appropriate [9, 29].

While there is comprehensive research on simulating AR environments in VR, to the best of our knowledge no prior research specifically investigated the simulation of projective AR in VR. Advances with respect to other AR modalities showed the potential of the general approach. It, therefore, seems plausible that simulating projective AR in VR can mitigate some of the AR modalities disadvantages. For instance, virtual projectors can be calibrated and

registered flawlessly with a virtual patient, which could facilitate the development and evaluation of future projective AR systems. However, distinct displaying properties of projective AR, e.g., distortion, reduced resolution and contrast as well as shadowing effects, may distinguish this technique from other displaying devices. Thus, related insights may not be as easily transferable. Therefore, this work investigated the potential of VR as a tool to facilitate the evaluation of medical projective AR visualizations and find answers to the following research question: *Is VR a feasible environment to evaluate medical projective AR visualizations?* We created a virtual twin of a real laboratory environment and implemented a first prototypical projective AR simulation. We then replicated a relevant user study in the field of evaluating medical projective AR visualizations, first in our own projective AR laboratory setup and then in our simulated virtual twin in VR. By showing that both studies yielded comparable results, we aimed to demonstrate the feasibility of conducting comparable experiments in VR.

2 TECHNICAL METHODS

To investigate whether VR provides a feasible environment for the evaluation of projective AR, we conducted an identical experiment in AR and in VR. For this initial experiment evaluating simulated projective VR, we specifically focused on a medical use case in the field of minimally-invasive interventions because of the promising advantages of projective AR in this domain [8, 21]. We based our study on one of the only articles comparing different projective AR visualizations in this application area. Heinrich et al. [13] conducted a comparative analysis of various visualization concepts to guide the insertion of a needle-shaped instrument. In this work, we aimed to replicate their experiment as closely as possible, both using our own laboratory setup and a VR simulation.

2.1 Visualization Concepts

The experiment of Heinrich et al. [13] compared three visualization concepts and two parametrization methods. To simplify our study and reduce the number of needed statistical tests, we only re-implemented the best and the worst-performing concepts of the original article. These included a crosshairs-shaped navigation concept and an arrow visualization. We implemented the techniques as described by Heinrich et al. [13]. In addition, no further parametrization was considered. We only implemented a continuous color mapping approach using a traffic lights metaphor and scaled every accuracy indicator linearly. The visualizations are explained in detail in Figure 2 and are also shown in Figure 1.

2.2 Physical AR Environment

As described by Heinrich et al. [13], the navigation concepts have been implemented in *Unity* (Unity Technologies, USA) as procedurally generated meshes at runtime. The visualizations were displayed using a Barco G60-W7 DLP laser phosphor projector (Barco GmbH, Karlsruhe, Germany). The system was calibrated using the photogrammetric measurement system *ProjectionTools*¹ (domeprojections.com GmbH, Berlin, Germany). This calibration also yielded a digital surface scan of the projection surface. For this, we used a human torso phantom containing ballistic gelatin in its center,

¹<https://www.domeprojection.com/software/projection-tools>

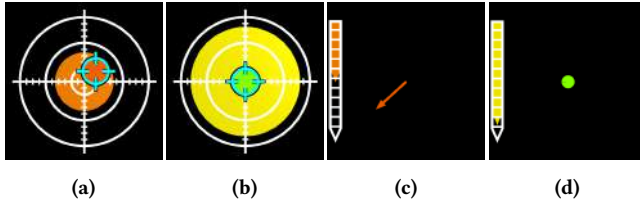


Figure 2: The needle navigation visualizations. (a) and (b) show the *Cross* concept. The small crosshairs (blue) need to align with the center of the larger crosshairs for optimal angular placement. The circular filling represents insertion depth. (c) and (d) show the *Arrow* concept. The arrow indicates the insertion trajectory and the progress bar communicates depth information. (a) and (c) as well as (b) and (d) show the identical insertion status, respectively.

allowing for realistic needle penetration. Using static projection mapping with the *domeprojection.com Unity Plugin*², AR contents were projected perspectively correctly onto the phantom. Needle position data was acquired with an infrared-light-based (IR) optical tracking system (spryTrack 180, Atracsys LLC, Switzerland) and fiducial markers attached to the used needle. Registration between the projection and the tracking system was achieved through the use of an optical IR marker with a known transformation relative to the known origin of the projection coordinate system. The complete experimental apparatus and laboratory can be seen in Figure 3a and Figure 4a.

2.3 VR Environment

The VR application was run on a computer equipped with a 3.4G Hz Intel(R) Core(TM) i7-6700 processor, 16.0 GB of RAM, and an NVIDIA GeForce GTX 1080 Ti graphics card. The VE was displayed through a Valve Index VR headset (Valve Corporation, Bellevue, WA, USA), which features two fully-resolving RGB-LCDs, each with a resolution of 1440x1600 pixels. The VR headset operated at a refresh rate of 120 Hz, contributing to increased realism and visual comfort within the virtual scene. To accurately replicate the real-world environment in VR, precise measurements of the dimensions and positions of fixed elements and the projector setup were conducted. To ensure an authentic representation, we recreated the physical space in Unity (version 2018.4.20f1) at a 1:1 scale. Virtual light sources aligning the position of the ceiling lights in physical space, mimicked the casting of shadows of virtual objects. Figure 3b illustrates the outcome of the VE and its individual components. The virtual representation of the user's hand was implemented as a rigid model. Participants were provided with blue surgical gloves to generalize the appearance of the hands, thus facilitating a better transfer to the virtual hand model. In VR, a second, slightly larger, semi-transparent hand model was superimposed, colored and textured to closely resemble the appearance of the real surgical glove. The representation of the user's hand in VR was necessary to enable a realistic simulation of projective AR, as the hand model generates a shadow when it occludes the projection surface. To achieve a realistic effect regarding intensity, shape, and size of the shadow, an

²<https://www.domeprojection.com/software/plugin-ins/unity-plugin>

additional spotlight source was implemented in the virtual scene at the projector's location, adjusted to match the physical projector's light cone. For realistic haptic sensations in VR, participants also had to insert a real needle into the same physical human torso phantom as in the AR condition. This can be seen in Figure 4b. The tracking data of the optical IR camera was used for the position of the virtual twin of the needle with the hand model being rigidly attached to it. We used the projection surface scan model obtained during projector calibration to represent the phantom in VR. The registration of the VE with the tracking system and the phantom was achieved using a HTC Vive tracker (HTC Corporation, Taoyuan, Taiwan) with a known transform to the AR world coordinate system.

2.4 Simulated projective AR

The implementation of our virtual projective AR simulation was based on the Unity package *Projector Simulator* [34], enabling the realistic projection of color images, image sequences, or videos. To more closely resemble the appearance of our laboratory projective AR setup, several steps were needed. Analogous to the AR application, visualizations for needle navigation were implemented as procedurally generated meshes in real-time. These visualizations were dynamically computed based on calculations between the surface of the phantom and the real-time needle tracking. The glyphs were rendered beyond the boundaries of the visible VE and captured by a virtual camera consistently centered above them. These renderings were then saved as a render texture, which was subsequently passed to the *Projector Simulator*. The virtual projector was positioned and oriented to match the projection direction of its physical counterpart. The brightness and color ratios of the simulated projector were manually adjusted within Unity to visually align as closely as possible with the real projection of the AR visualizations.

3 EVALUATION METHODS

This work's general objective is to assess whether VR is a feasible environment to evaluate medical projective AR visualizations. To this end, a mixed design user study was conducted to compare two needle navigation aids. The visualizations were evaluated as a within factor. This comparison was then conducted both using conventional projective AR and using simulated AR in VR in a between-subjects design manner. The mediation modality, i.e., AR or VR, was the primary factor of this investigation.

3.1 Hypotheses

This section details the hypotheses investigated in this evaluation.

3.1.1 Performance Measures. The conventional projective AR comparison was based on a study by Heinrich et al. [13]. This study included three objective user performance indicators in the form of angular and depth setting accuracy, as well as task completion times. To consider VR a feasible environment to evaluate these visualizations, similar experimental outcomes would need to be observed, i.e., the same method would need to show superior performance measures in both modalities. Therefore, the first hypothesis was: *H1: There are no differences in means between the AR and the VR*



Figure 3: The experimental apparatus. A projector and an IR-based optical tracking camera were used to navigate the insertion of a needle into a gelatin filled phantom. (a) shows the physical laboratory and (b) depicts its virtual twin.

group regarding relative differences of objective performance measures between both visualization concepts.

3.1.2 Subjective Workload. Besides investigating performance data, subjective workload and mental demand were also considered crucial criteria. In order to be able to compare visualization methods in VR, it was important that the relative difference between navigation approaches was similar in AR and in VR. This resulted in the following hypothesis: *H2: There are no differences in means between the AR and the VR group regarding relative differences in the subjective workload evoked by visualization concepts.*

3.2 Sample Design

Participants with a general medical background were considered for the study. Our research objectives did not necessitate specific clinical experience. Therefore, we extended invitations to students pursuing degrees in both human medicine or medical engineering at the authors institution. In total, 43 students were recruited for the study. 22 (16 female, 6 male) of the participants were randomly assigned to the AR group, 21 (16 female, 5 male) to the VR group. In the AR group, 20 participants studied human medicine and two studied medical engineering. The participant age ranged from 21 years to 37 years old (median: 24 years) and participants were between their fourth and 12th year of university. In the VR group, 19 medical students and two medical engineering students participated. The participant age ranged from 20 years to 31 years old (median: 24 years) and participants were between their second and 12th year of university. None of the students who participated in the study reported any degree of deuteranopia.

3.3 Tasks

For each presented visualization concept, participants were asked to complete a needle insertion task. They were instructed to insert a tracked needle-shaped instrument into a phantom filled with ballistic gelatin as accurately as possible by keeping the predefined insertion angle and reaching a predefined insertion depth without over-inserting the needle. The needle insertion specifications were communicated by a projected navigation visualization. Participants

were always instructed to favor accuracy over time. Figure 4 shows the execution of the task in AR and in VR.

3.4 Variables

The hypothesis *H1* is directed towards objective performance indicators of the needle insertion task. Based on the original study by Heinrich et al. [13], three measures were selected and obtained accordingly. First, angular accuracy, i.e., how well the planned insertion angle was set, was measured by calculating the angle between the tracked needle and the desired insertion path. This was repeated every 3.3 mm the instrument was advanced. The mean value of all sample points was then regarded as the *average angle error* throughout the insertion process. This calculation excluded all data points obtained in the first 2 s after the trial started and during the first 1 cm of needle insertion to attenuate larger errors during initial angle adjustments. Second, depth accuracy, i.e., how well the planned insertion depth was set, was measured as the difference between the desired insertion depth and the Euclidean distance between the needle tip and the needle entry site. The absolute value of this difference was considered as the *absolute distance error*. Third, *task completion time* was measured as the time passed between a starting signal given by the experimenter and a stopping signal given by the participant. Moreover, the raw version of the standardized NASA



Figure 4: The experimental task. A user is inserting a tracked needle using projective AR guidance (a) and using simulated AR guidance in VR (b). The non-dominant hand was held at the back to prevent the risk of injuries.

Table 1: Summary of descriptive results for: AAE - average angle error (°), ADE - absolute distance error (mm), TCT - task completion time (s), and SWL - subjective workload. All entries are in the format: mean value [standard deviation].

	AAE	ADE	TCT	SWL
Total	1.21 [1.09]	1.33 [0.77]	40.52 [17.37]	35.03 [15.43]
Arrow	1.07 [0.99]	1.75 [0.86]	44.06 [15.98]	40.45 [15.83]
Cross	1.35 [1.18]	0.91 [0.33]	36.98 [18.14]	29.61 [13.09]
AR	1.41 [1.35]	1.35 [0.69]	40.87 [21.82]	33.39 [16.12]
Arrow	1.13 [1.06]	1.79 [0.66]	44.19 [20.10]	38.14 [17.67]
Cross	1.69 [1.56]	0.91 [0.35]	37.54 [23.41]	28.64 [13.13]
VR	1.00 [0.68]	1.31 [0.86]	40.15 [11.20]	36.75 [14.68]
Arrow	1.01 [0.93]	1.71 [1.04]	43.92 [10.60]	42.86 [13.66]
Cross	0.99 [0.29]	0.91 [0.32]	36.39 [10.73]	30.63 [13.30]

task load index [11] was used to assess *H2*. The mean value of all items was considered as the overall *subjective workload* on a scale from 0 to 100.

3.5 Procedure

The study commenced with a short introduction, a collection of informed consent and a query of participant data. Afterwards, the experimental setup and equipment, as well as the tasks and the procedure were explained in detail. We then asked the participants to put on blue surgical gloves independently of whether they were testing the AR or VR application. The participants testing the VR application were additionally equipped with a VR headset. Subsequently, the first visualization concept was explained and a training phase began. Training was conducted in two parts. The first part (first two training trials) served to learn and familiarize with the respective visualization. The second part (third training trial) was conducted under experimental conditions. If no open questions remained, three consecutive trials with the current visualization concept were conducted. Target insertion angle and depth were generated randomly before each trial. Ranges resulting in comparable task difficulty were selected for this. Insertion depths could range between 70 mm and 90 mm. Insertion angles were selected between 0° and 30° at a randomized direction. Each trial was started by the investigator and ended with a stop command by the participants when they believed to have optimally reached the target. After those three trials, participants were asked to complete the NASA-TLX questionnaire for the first visualization concept. Then, the whole process was repeated with the respective other visualization concept. The order of visualization concepts was alternated for each participant.

4 RESULTS

4.1 Data Preparation

After collecting the experimental raw data, mean values of the repeated measures of identical experimental conditions were calculated for each participant, e.g., all task completion time data using the *Arrow* concept of a participant were aggregated into a single value. These experimental outcomes are summarized in Table 1.

Complete NASA TLX results are additionally reported in the supplementary Table 3. In this study, we did not pursue the goal that visualization conditions perform equivalently between AR and VR. We were rather interested in whether comparable differences between navigation visualizations could be observed in AR and in VR. To this end, for each participant, a ratio score between the *Arrow* and the *Cross* data was calculated by dividing an *Arrow* value by the respective *Cross* value. Following Hedges et al. [12], the natural logarithms of these ratios were then calculated to linearize the data and to facilitate a more normally distributed sample. Resulting positive scores then indicated higher *Arrow* concept measures while negative scores correlated to higher *Cross* concept values.

4.2 Statistical Analyses

All statistical analyses to test the hypotheses *H1* and *H2* were conducted using R (version 4.1.1). First, Q-Q plots were calculated to ensure normality of the data. Afterwards, t-tests were conducted to determine if there were significant differences between the AR and the VR group. In case no significant effects were found, Bayes factors were calculated to estimate how well the null hypotheses, that there are no differences between groups, could be supported [16]. The Bayes factor results were interpreted according to the thresholds defined by Lee and Wagenmakers [24]. All statistical results are summarized in Table 2 and visualized in Figure 5.

4.3 Interpretation of Results

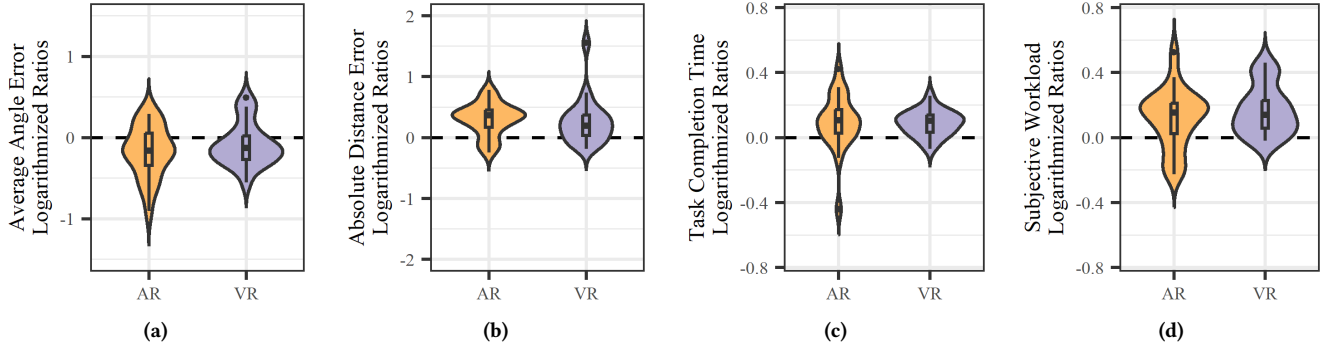
Regarding *H1*, no significant differences could be found between the logarithmized ratios of performance measures between the AR and the VR group. Figure 5a - Figure 5c show that the distributions and medians of relative differences were similar for both study groups. Bayes factor analyses showed that there is moderate evidence favoring the null hypotheses of the conducted statistical tests, that there are no differences between the AR and the VR group regarding the absolute distance error and task completion time. The respective Bayes factor for the average angle error is also in favor of the null hypothesis, however only to an anecdotal extent. Overall, *H1* can, thus, be accepted. Regarding subjective workload, the logarithmized ratios between the *Arrow* and the *Cross* concepts showed comparable distributions for both groups with similar median scores (see Figure 5d). This means, that in both modalities the *Cross* concept was rated as less demanding compared to the *Arrow* concept and that relative difference between those ratings was on a comparable level. This finding is further supported by the non-significant nature of the conducted t-test and a Bayes factor supporting anecdotal evidence in favor of the test's null hypothesis, that there are no differences between the AR and the VR group. Thus, *H2* can at least partly be accepted.

5 DISCUSSION & CONCLUSION

This work tried to find answers to the question: *Is VR a feasible environment to evaluate medical projective AR visualizations?* Statistical analyses revealed no significant differences between both study outcomes. On the contrary, Bayes factor analyses showed that there is anecdotal to moderate evidence supporting the hypotheses that both the VR group and the AR group stem from the same population, i.e., that both studies yielded comparable results.

Table 2: Summary of statistical analyses on dependent variables.

Variable	<i>df</i>	<i>t</i>	<i>p</i>	Cohen's <i>d</i>	Effect	<i>BF</i> ₁₀	Evidence for <i>H</i> ₀	Figure
Average angle error	40.275	-0.917	0.365	0.278	small	0.419	anecdotal	Figure 5a
Absolute distance error	35.554	0.406	0.687	0.125	-	0.321	moderate	Figure 5b
Task completion time	29.243	0.126	0.901	0.038	-	0.302	moderate	Figure 5c
Subjective workload	40.079	-1.038	0.306	0.315	small	0.460	anecdotal	Figure 5d

**Figure 5: Violin charts of the experimental results. Plots show the logarithmized ratio between the *Arrow* and the *Cross* concepts regarding a) average angle errors, b) absolute distance errors, c) task completion time, and d) subjective workload data.**

Therefore, the raised research question can be cautiously affirmed. With respect to navigated needle insertion tasks, VR occurs to be a feasible environment to evaluate medical projective AR visualizations. However, these findings can only be applied carefully to other experiments. Future work is needed to replicate similar studies as reported in this work to further consolidate VR as a meaningful alternative to physical projective AR prototypes. Besides that, our study was influenced by several limitations that should be avoided or investigated in future work.

In our experiment, we decided on a rigid hand model for virtual representation of the user's hand because of occlusion problems while tracking the hand holding the needle. Real-time hand tracking may have evoked a greater sense of embodiment and virtual body ownership, thus potentially improving the overall user experience immersion in VR. However, due to the rather static position of the participants' real hands while handling the needle, we hypothesize that enabling hand tracking would not have affected our results significantly. Moreover, the calculation of the projective AR representation in VR differed from the physical projective AR representation due to limitations of the simulation approach compared to projection mapping used in the AR condition. Implementing a high-fidelity virtual projection system and evaluating its resemblance to conventional projective AR was not within the scope of this work. Instead, we were more interested in first assessing the general feasibility of the simulation to produce comparable study results. Besides the projection itself, multiple other fidelity aspects were considered in this work, e.g., the general design of the VE. Future work could investigate effects of different fidelity levels to find optimal compromises between implementation effort and outcomes, similar to the work of Schott et al. [33].

In addition, the tested VR condition in our study still required a physical mock-up to simulate the needle insertion. For even faster

prototyping and evaluations, it may be beneficial, to only insert needles virtually. However, it remains uncertain, how such interaction would affect the outcomes of comparable experiments. Finally, this work only compared logarithmized ratio data of the AR and the VR group. This was done because we wanted to show that both studies can yield comparable results, i.e., that in both experiments the same visualization performed best. We deliberately decided against comparing the raw outcomes summarized in Table 1, as we did not aim for this initial evaluation to show that, for instance, the *Arrow* concept performs equivalently in AR and VR. Rather, we expected differences between both mediation conditions, but hypothesized that the general trends between compared visualization concepts would be comparable. Hence, we wanted to show that VR can be used to find out which visualization method generally performs best, but assumed that using this modality no reliable predictions regarding absolute accuracy values in real AR scenarios are feasible. Future studies should investigate this aspect in more detail.

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A SUBJECTIVE WORKLOAD DATA

This section contains the complete descriptive data collected using the NASA TLX questionnaire (see Table 3). Data of the six sub-scales mental, physical, and temporal demand, as well as performance effort and frustration are reported. Each item was rated on a scale from 0 to 20. Raw TLX scores listed in Table 1 represent the average of the six sub-scales multiplied by 5.

Table 3: Summary of descriptive NASA TLX results for. All entries are in the format: mean value [standard deviation].

	Mental Demand	Physical Demand	Temporal Demand	Performance	Effort	Frustration
Total	8.70 [4.92]	7.01 [4.23]	5.95 [4.34]	5.76 [3.99]	9.01 [4.52]	5.60 [4.28]
Arrow	9.77 [4.90]	7.72 [4.37]	6.58 [4.74]	6.60 [4.32]	10.79 [4.11]	7.07 [4.65]
Cross	7.63 [4.76]	6.30 [4.02]	5.33 [3.86]	4.91 [3.46]	7.23 [4.25]	4.14 [3.33]
AR	8.50 [5.10]	6.66 [4.19]	5.89 [4.15]	5.45 [3.61]	8.07 [4.60]	5.50 [4.44]
Arrow	9.45 [5.23]	7.23 [4.65]	6.41 [4.48]	6.32 [4.04]	9.77 [4.65]	6.59 [5.15]
Cross	7.55 [4.89]	6.09 [3.70]	5.36 [3.82]	4.59 [2.95]	6.36 [3.96]	4.41 [3.38]
VR	8.90 [4.78]	7.38 [4.29]	6.02 [4.59]	6.07 [4.37]	10.00 [4.27]	5.71 [4.15]
Arrow	10.10 [4.62]	8.24 [4.10]	6.76 [5.11]	6.90 [4.68]	11.86 [3.23]	7.57 [4.12]
Cross	7.71 [4.73]	6.52 [4.41]	5.29 [3.99]	5.24 [3.97]	8.14 [4.44]	3.86 [3.34]